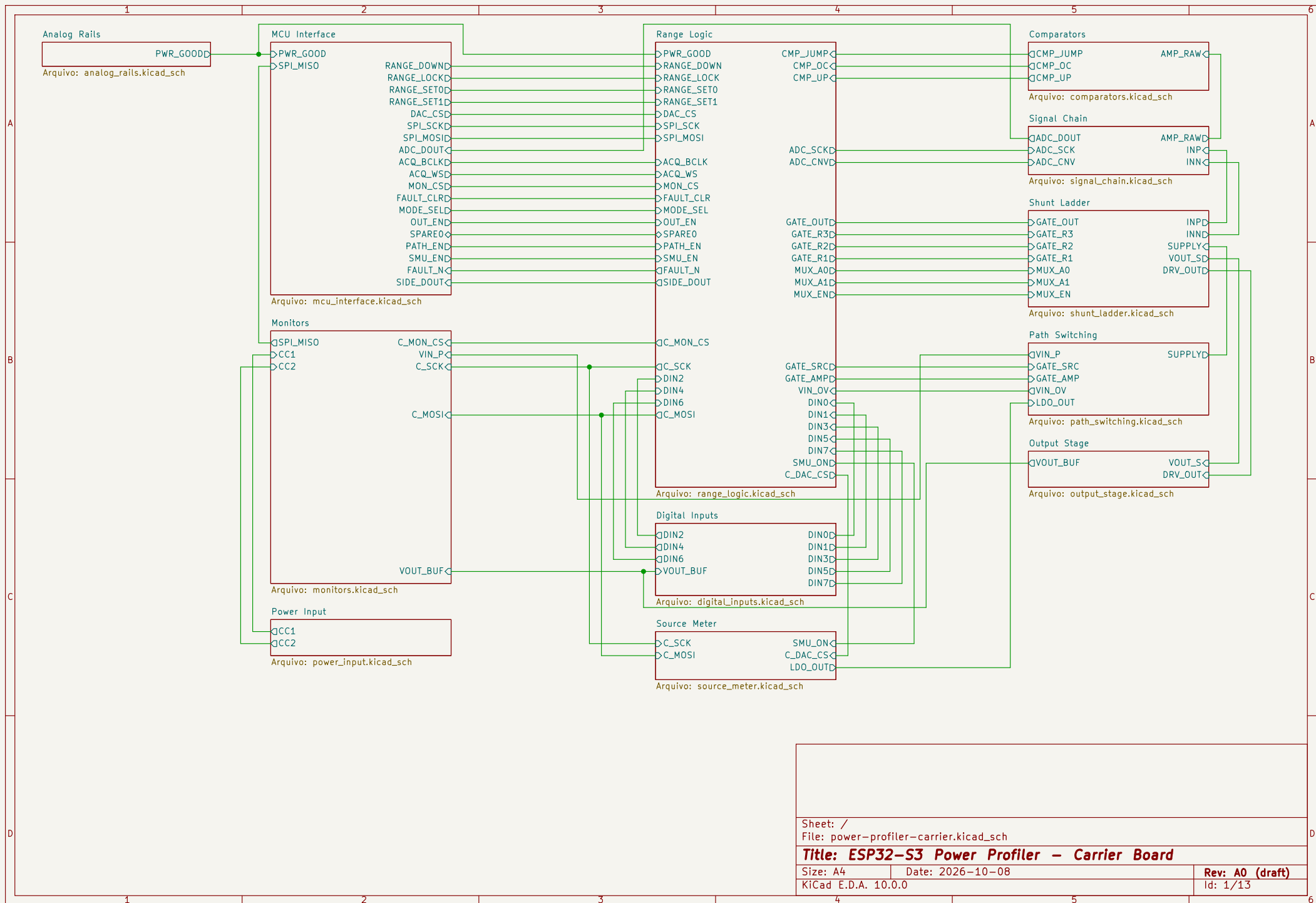
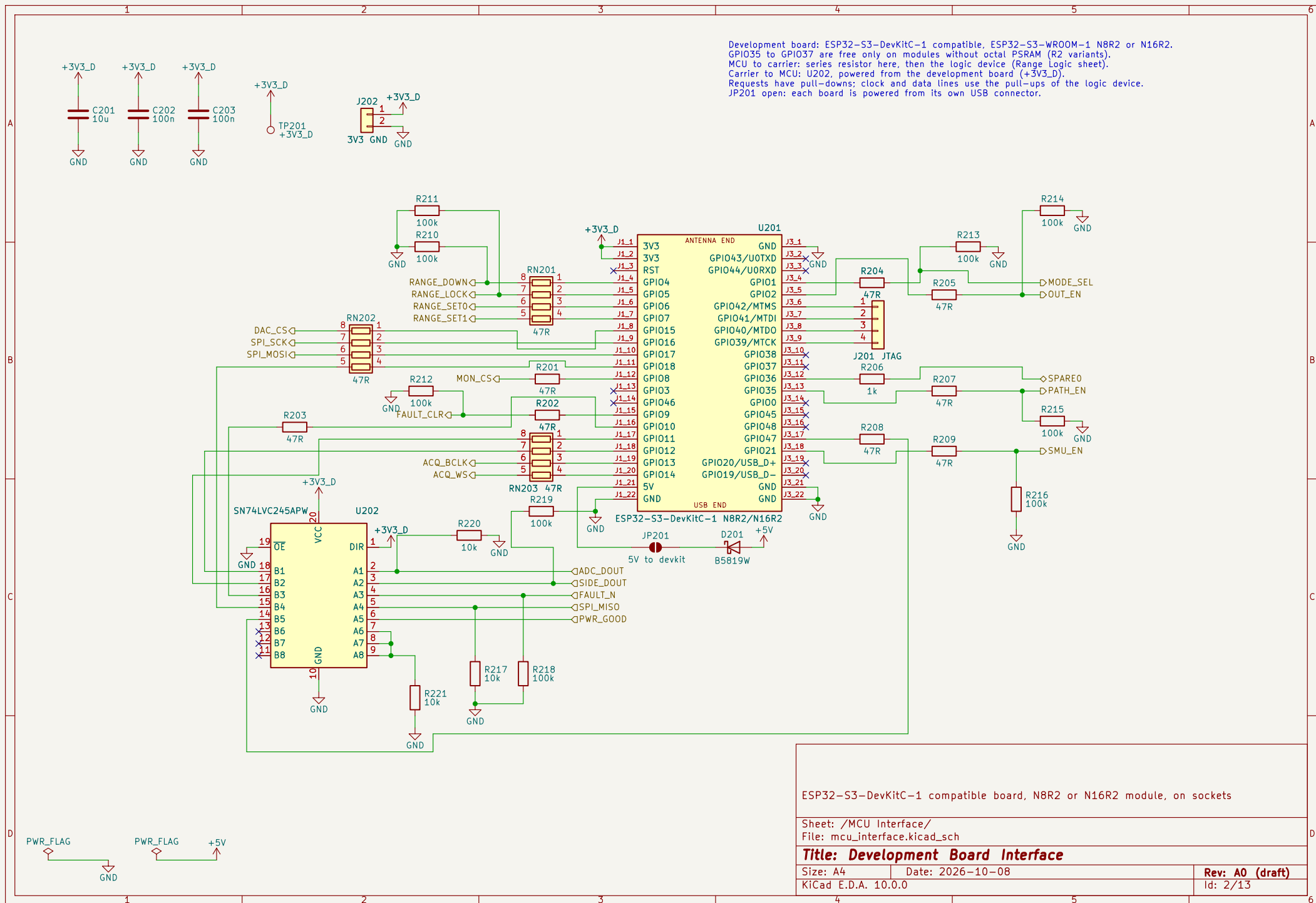






ESP32-S3-DevKitC-1 compatible board, N8R2 or N16R2 module, on sockets

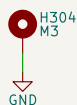
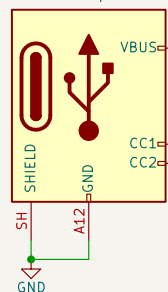
Sheet: /MCU Interface/
 File: mcu_interface.kicad_sch

Title: Development Board Interface

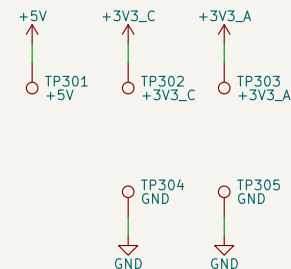
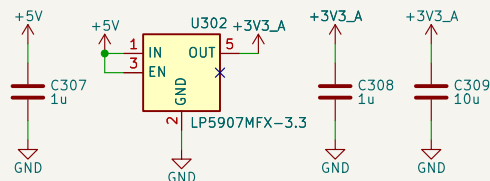
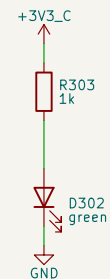
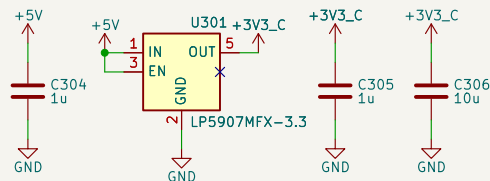
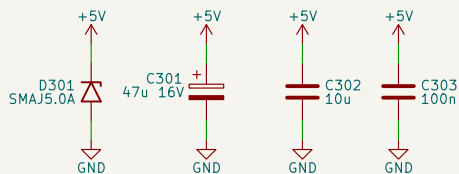
Size: A4 Date: 2026-10-08
 KiCad E.D.A. 10.0.0

Rev: A0 (draft)
 Id: 2/13

J301
USB-C power



Source capability from the CC voltage (firmware, through the monitor converter):
0.25 V to 0.61 V default USB, 0.70 V to 1.16 V 1.5 A, 1.31 V to 2.04 V 3 A.
+3V3_C: logic of the carrier. +3V3_A: converter, its driver and the comparators.
H301 to H304: mounting holes, tied to ground.



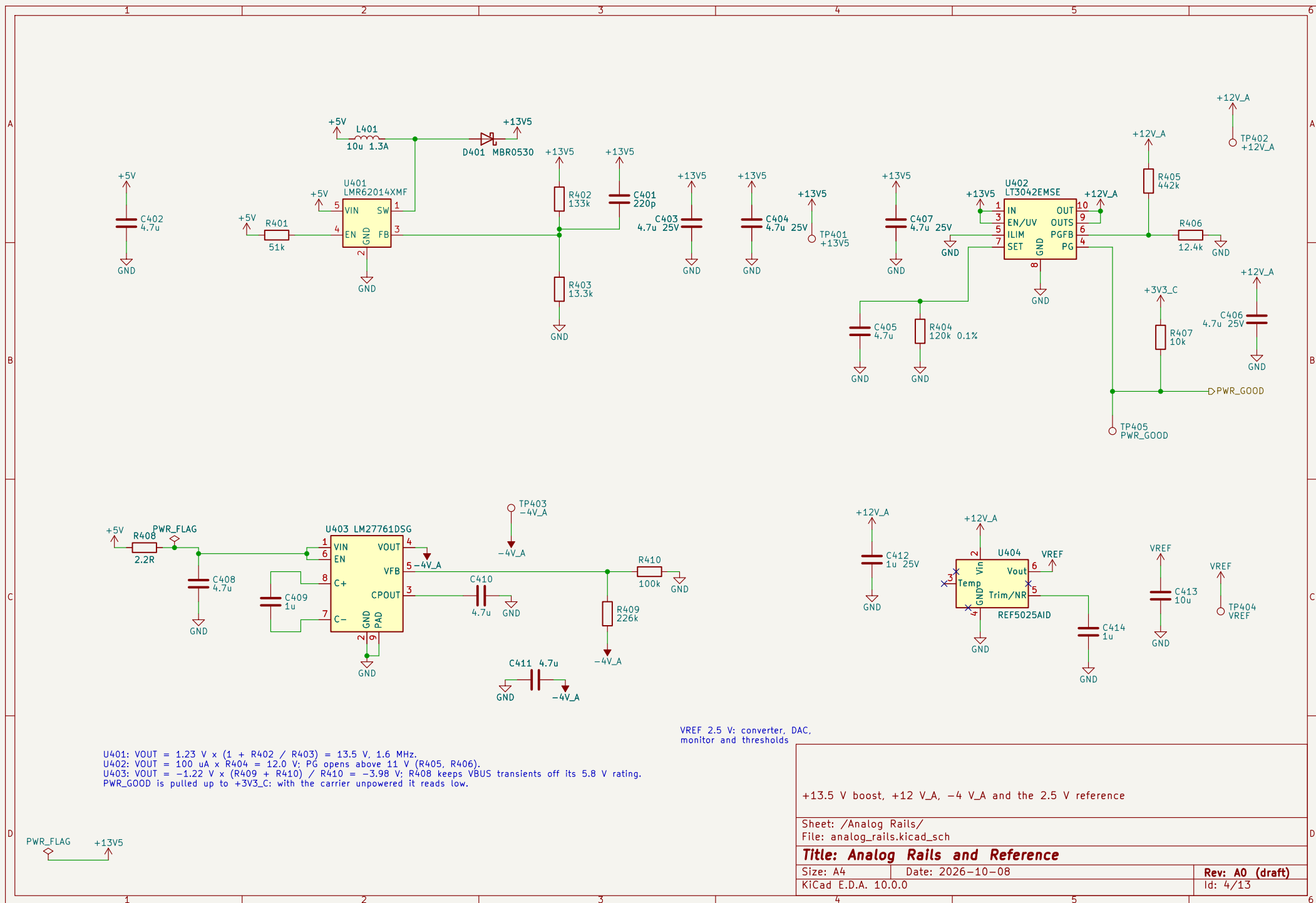
USB-C sink, 5 V only; CC voltages go to the monitor converter

Sheet: /Power Input/
File: power_input.kicad_sch

Title: Power Input and Logic Supplies

Size: A4 Date: 2026-10-08
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Id: 3/13



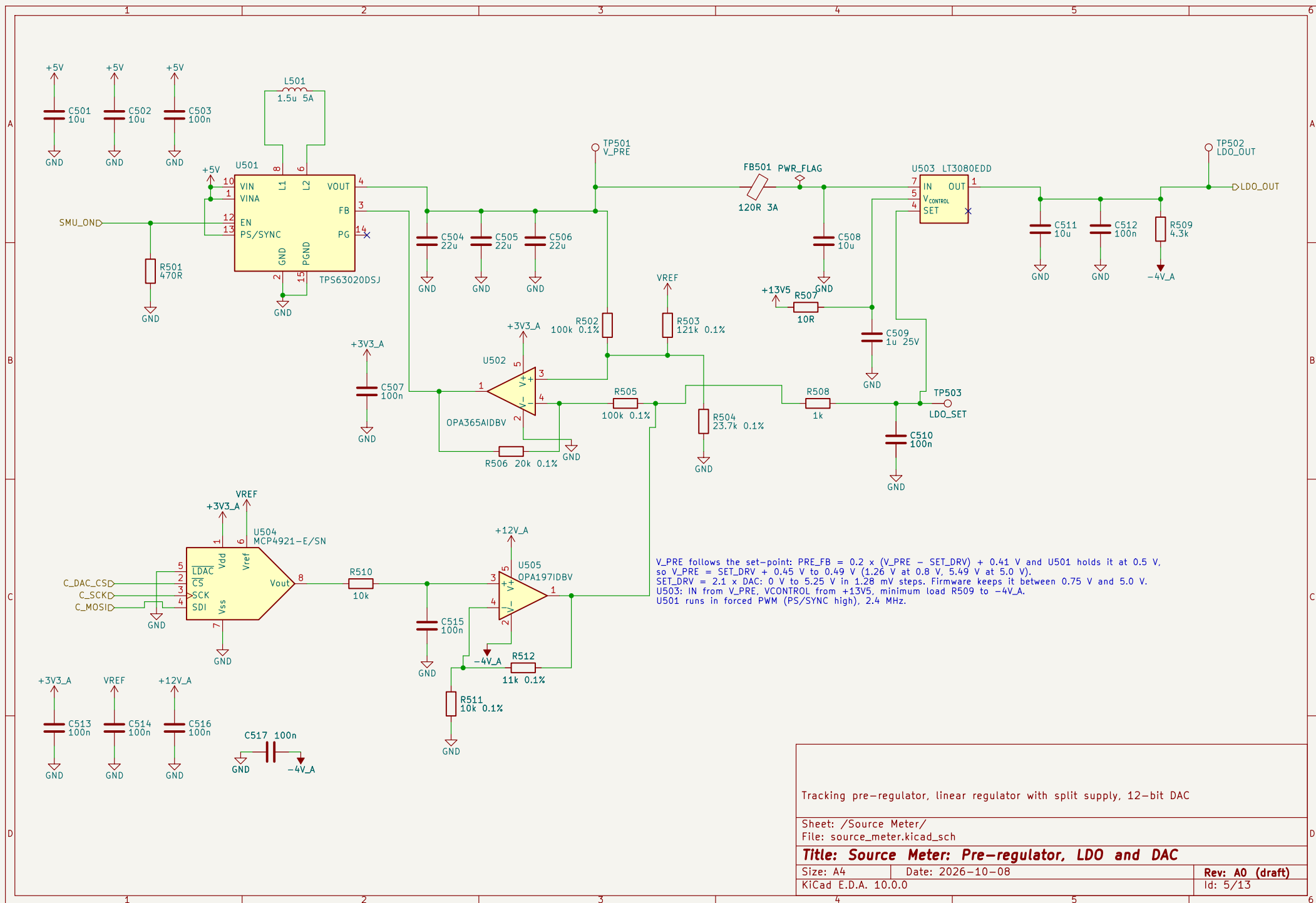
+13.5 V boost, +12 V_A, -4 V_A and the 2.5 V reference

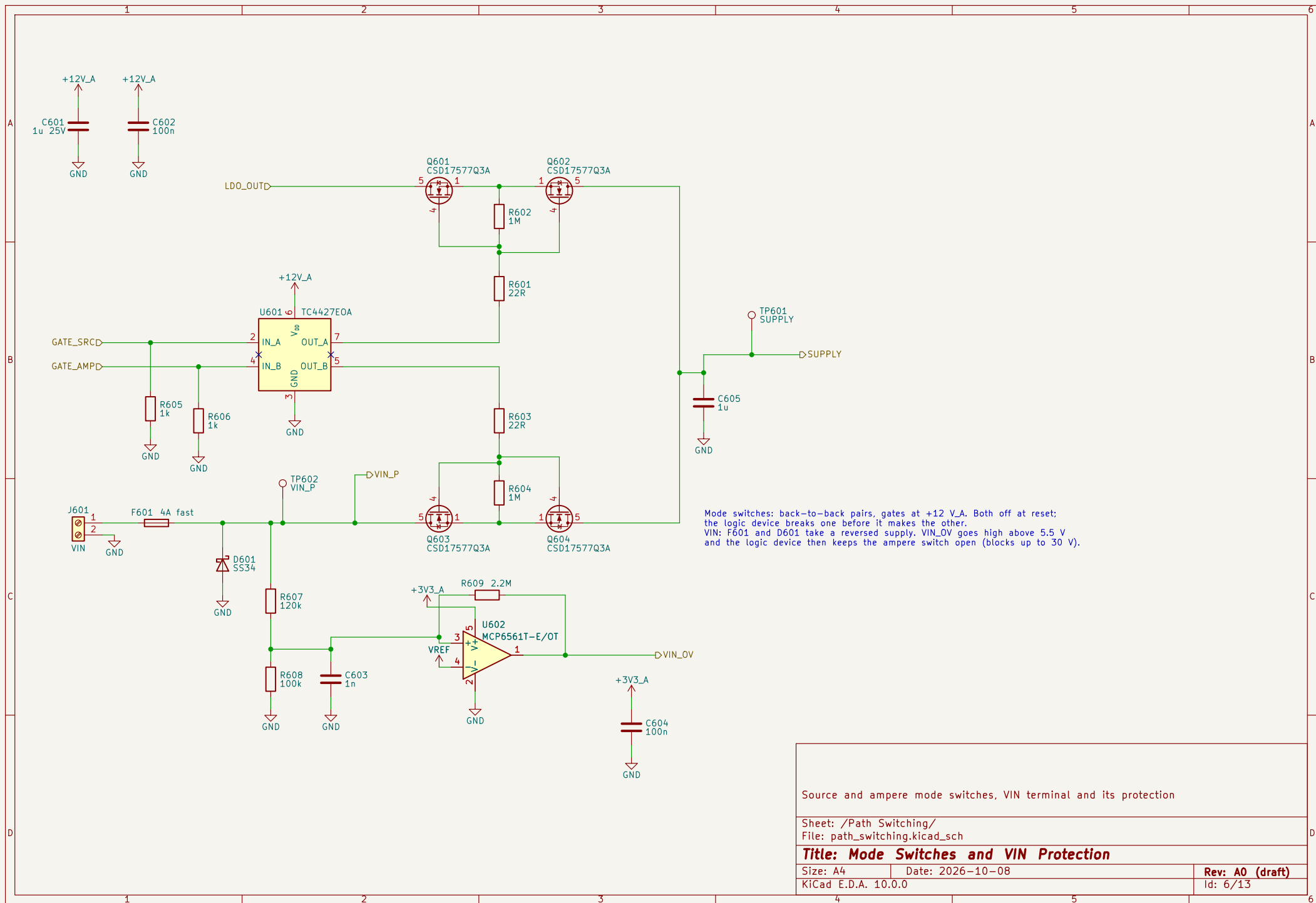
Sheet: /Analog Rails/
File: analog_rails.kicad_sch

Title: Analog Rails and Reference

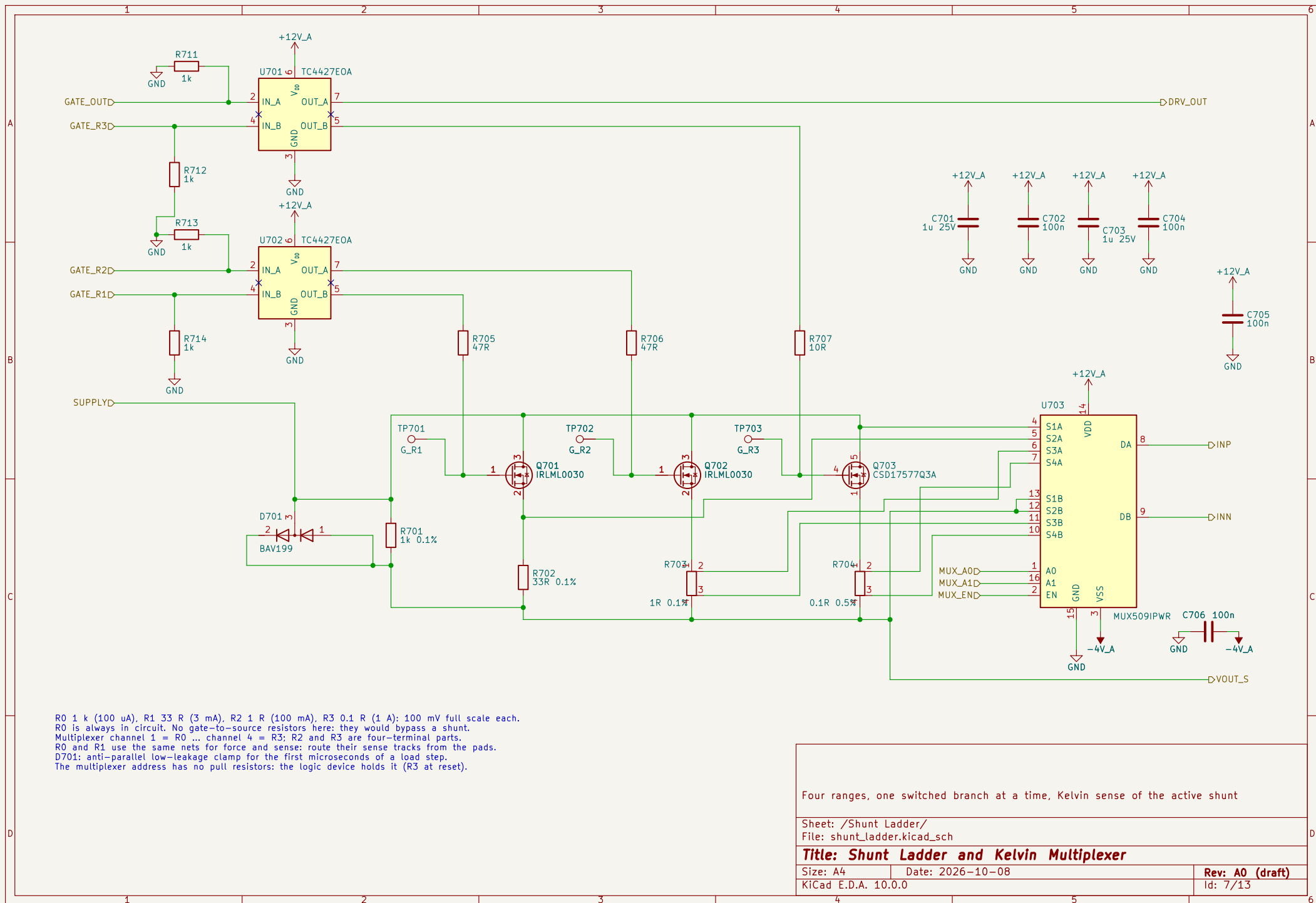
Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.0

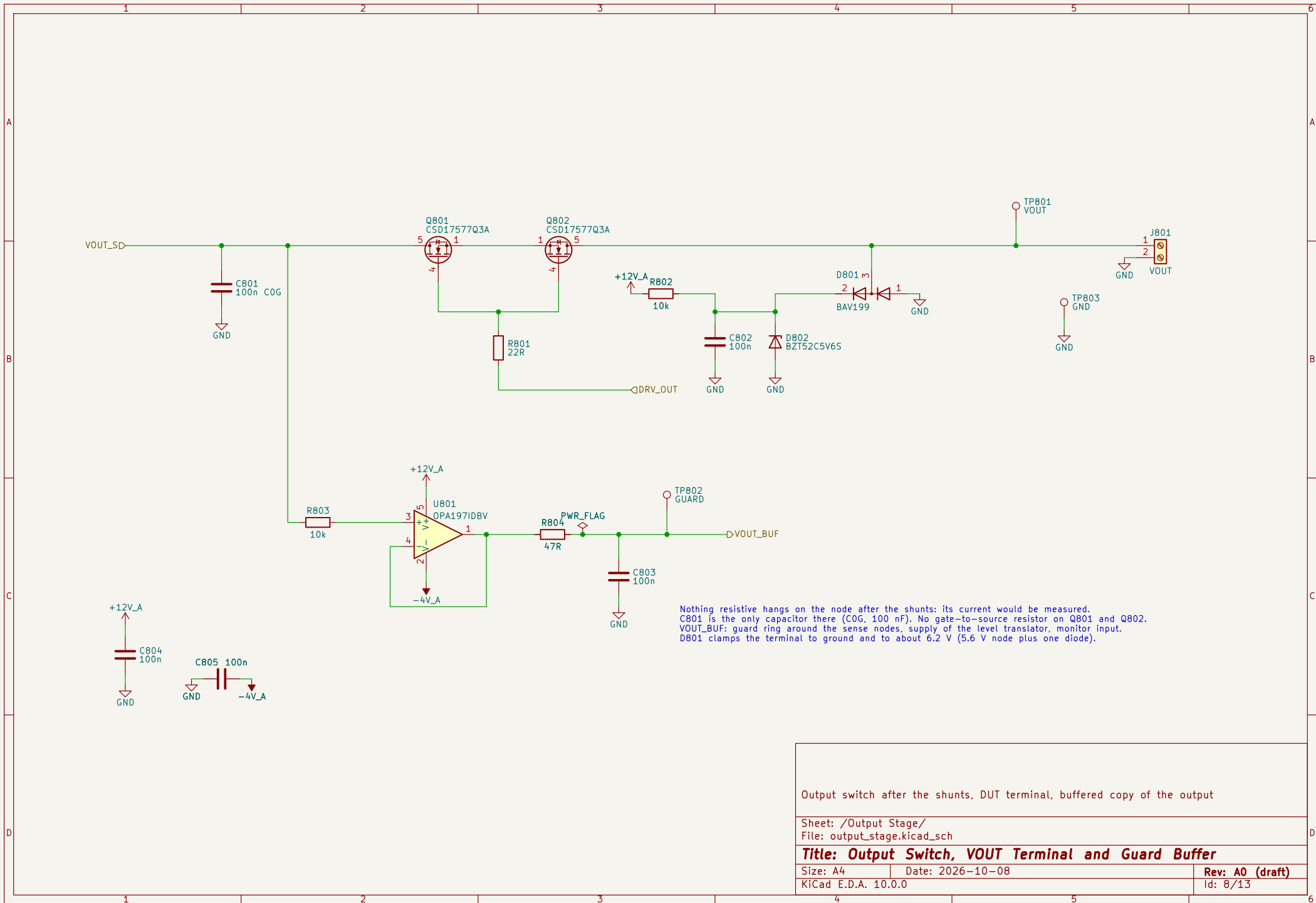
Rev: A0 (draft)
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Source and ampere mode switches, VIN terminal and its protection		
Sheet: /Path Switching/ File: path_switching.kicad_sch		
Title: Mode Switches and VIN Protection		
Size: A4	Date: 2026-10-08	Rev: A0 (draft)
KiCad E.D.A. 10.0.0	Id: 6/13	





Output switch after the shunts, DUT terminal, buffered copy of the output

Sheet: /Output Stage/
 File: output_stage.kicad_sch

Title: Output Switch, VOUT Terminal and Guard Buffer

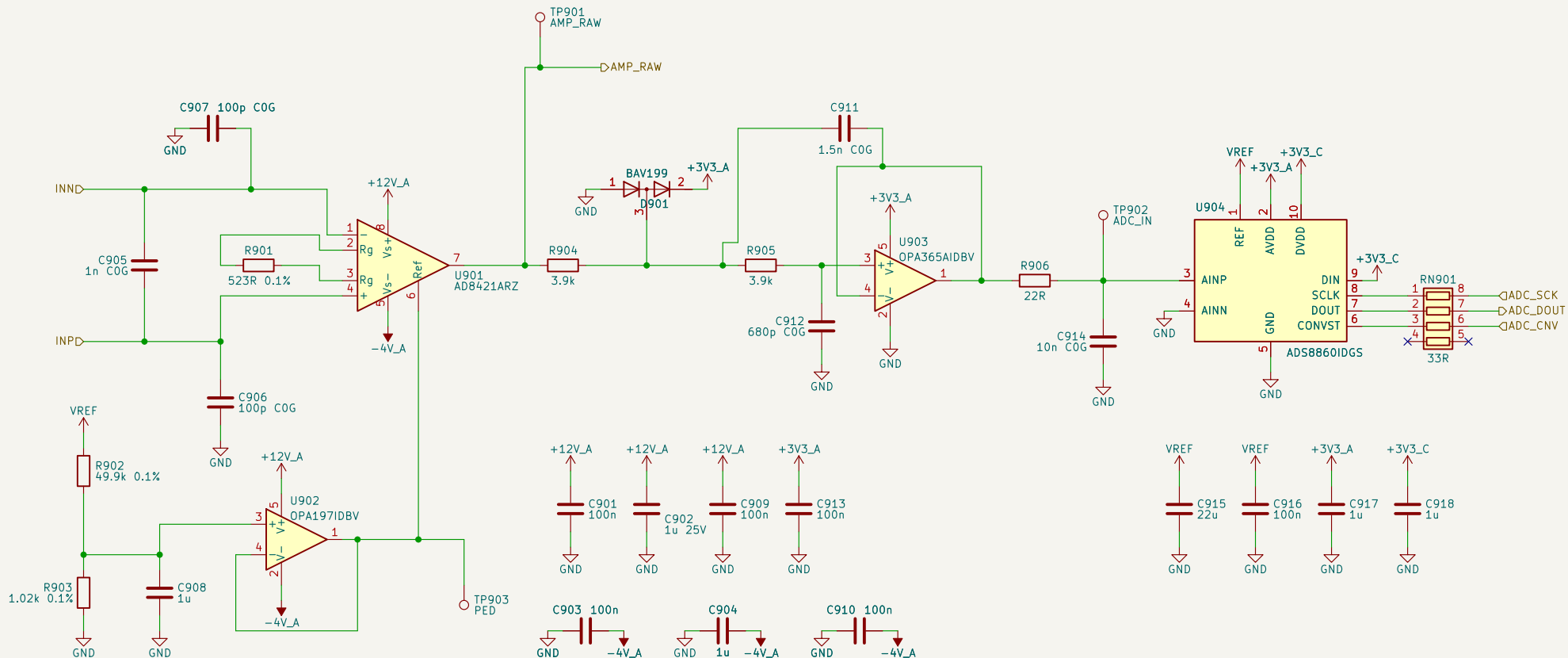
Size: A4

Date: 2026-10-08

Rev: A0 (draft)

KiCad E.D.A. 10.0.0

Id: 8/13



U901: $G = 1 + 9.9 \text{ k} / R901 = 19.93$. Reference pin at 50 mV; zero current reads about 1300 codes.
 AMP_RAW can reach 10 V during a range change; R904 and D901 limit the driver input.
 U903: Sallen-Key, 40 kHz, $Q = 0.74$. R906 and C914 as the converter input needs.
 U904: 3-wire mode, DIN high. CONVST is the I2S word clock, SCLK the bit clock.
 C915 and C916 sit at the REF pin.

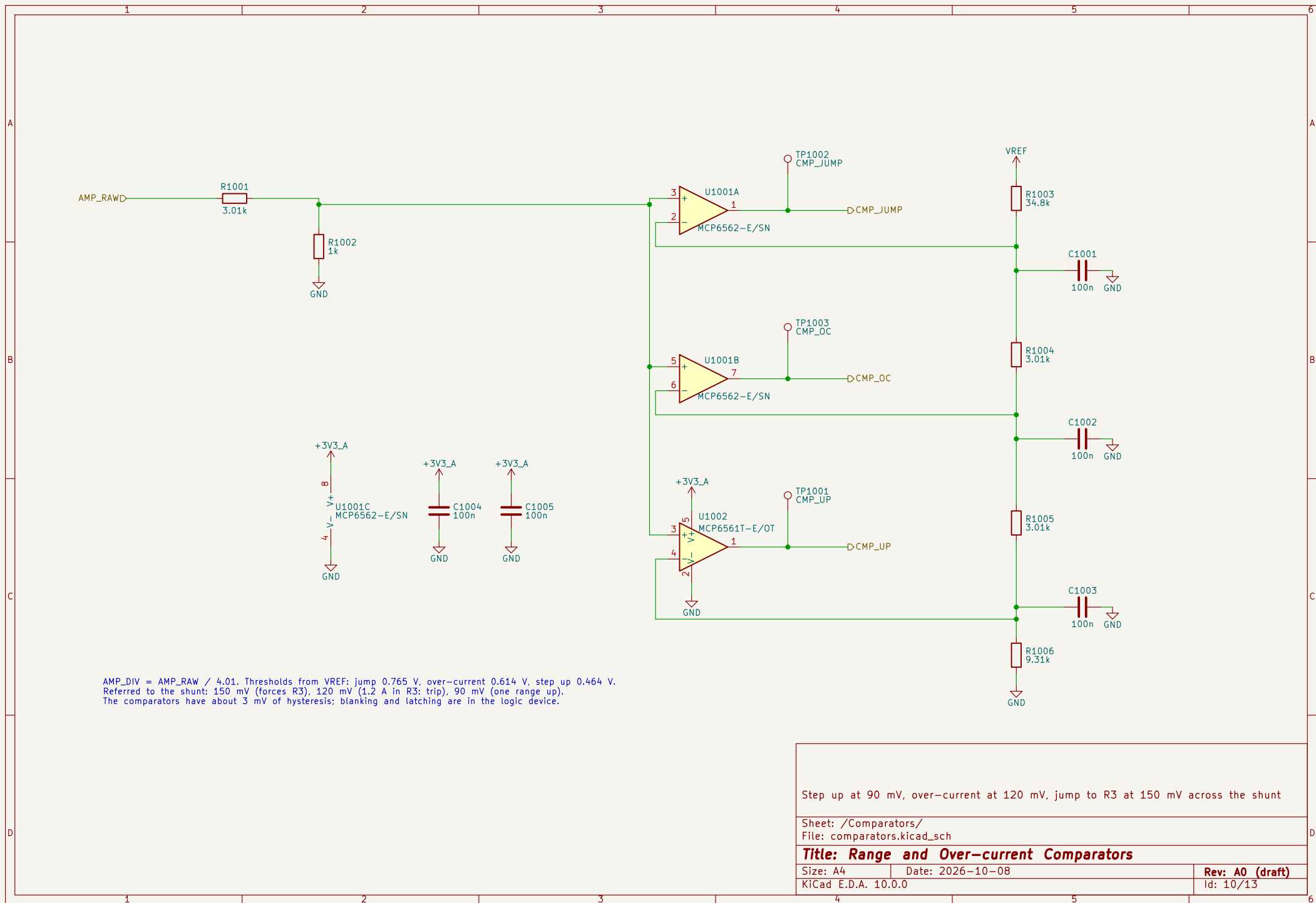
Gain 19.93, 50 mV pedestal, limiter, 40 kHz two-pole filter, 16-bit converter

Sheet: /Signal Chain/
 File: signal_chain.kicad_sch

Title: Amplifier, Anti-alias Filter and ADC

Size: A4 Date: 2026-10-08
 KiCad E.D.A. 10.0.0

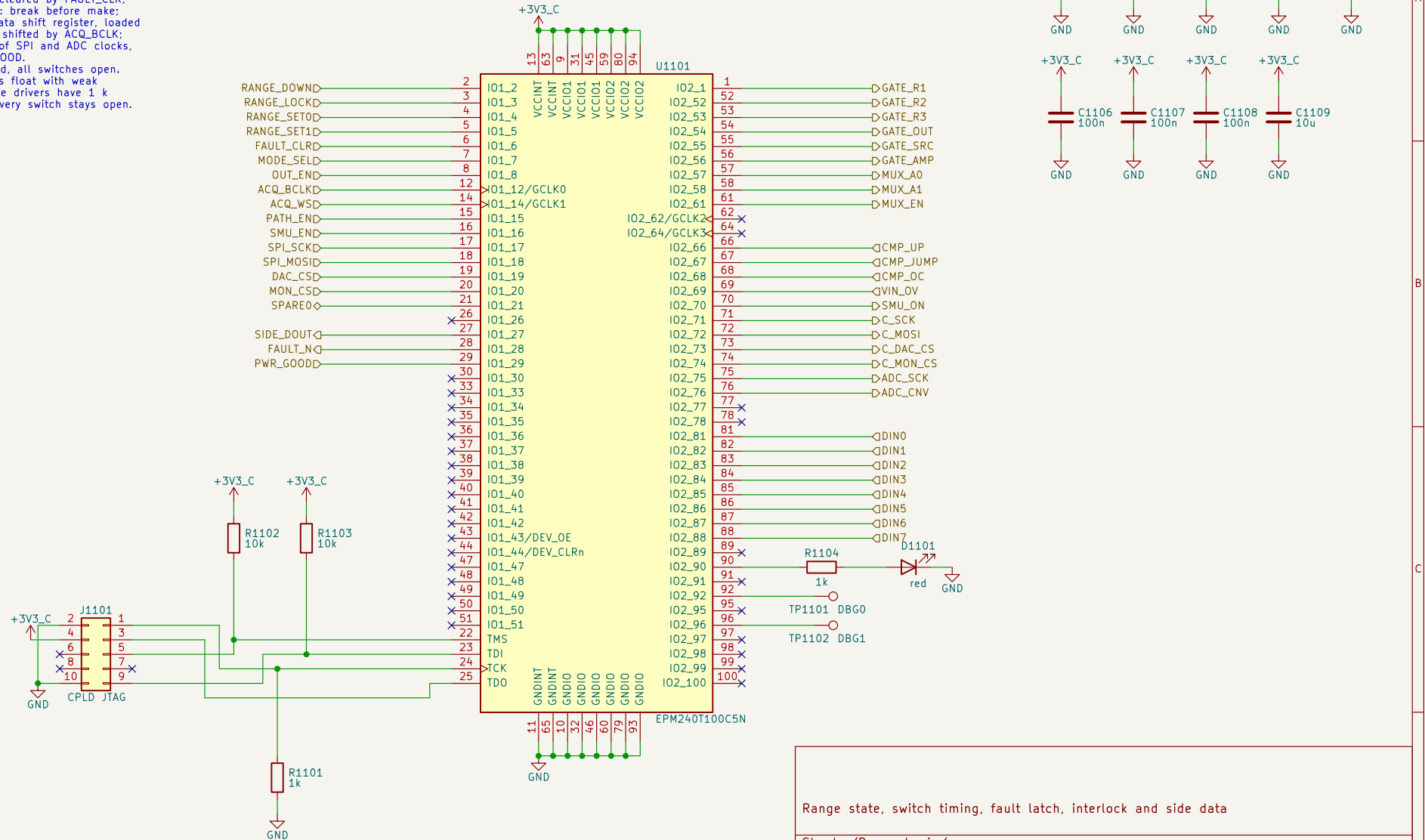
Rev: A0 (draft)
 Id: 9/13



U1101 holds, in HDL:

- range register R0..R3, step up and jump from the comparators, step down on a RANGE_DOWN pulse, lock to RANGE_SET;
- one-hot gate outputs with about 1 us of make-before-break, multiplexer address;
- fault latch (over-current, VIN_OV, loss of PWR_GOOD), cleared by FAULT_CLR;
- mode interlock: break before make;
- 16-bit side-data shift register, loaded by ACQ_WS and shifted by ACQ_BCLK;
- pass-through of SPI and ADC clocks, gated by PWR_GOOD.

Reset: R3 selected, all switches open.
Blank device: pins float with weak pull-ups; the gate drivers have 1 k pull-downs, so every switch stays open.



Range state, switch timing, fault latch, interlock and side data

Sheet: /Range Logic/
File: range_logic.kicad_sch

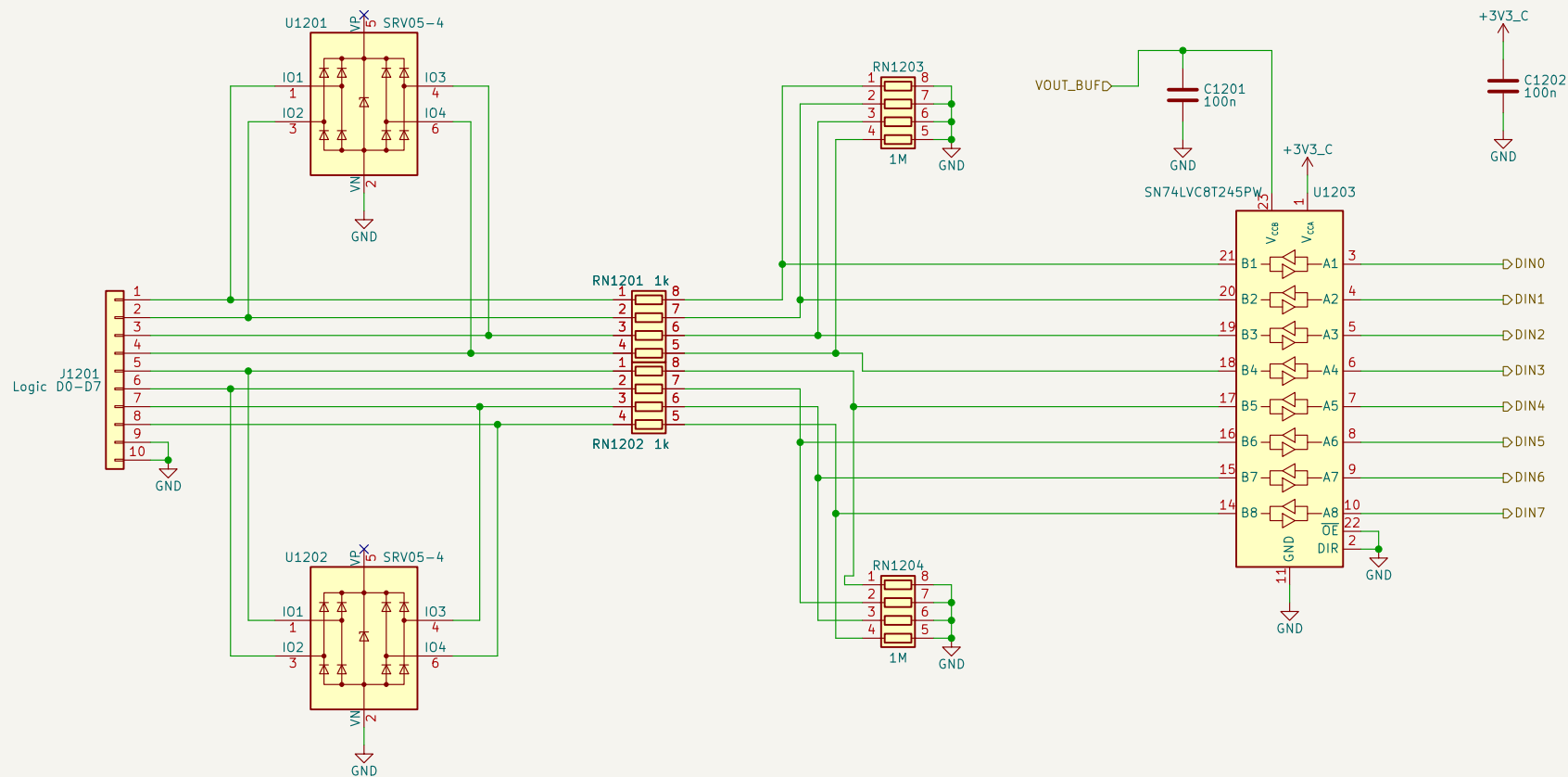
Title: Range Logic (CPLD)

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.0

Rev: A0 (draft)

Id: 11/13



J1201 is a 2x5 header: pins 1 to 8 are D0 to D7, pins 9 and 10 ground.
 U1203 port B is powered from VOUT_BUF, so the inputs follow DUT logic levels from 1.65 V to 5.5 V.
 DIR and OE are referenced to port A (+3V3_C): DIR low sets the direction B to A.
 With VOUT_BUF off the outputs float; the logic device holds DIN0 to DIN7 with its pull-ups.
 Each input loads the DUT pin with 1 M to ground.

Logic inputs follow the DUT supply through the buffered output voltage

Sheet: /Digital Inputs/
 File: digital_inputs.kicad_sch

Title: Digital Inputs D0 to D7

Size: A4	Date: 2026-10-08	Rev: A0 (draft)
KiCad E.D.A. 10.0.0		Id: 12/13

